Parkinson's disease detection using Handwriting

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Abstract— **Parkinson's Disease (PD) is a progressive neurodegenerative disorder characterized by the deterioration of dopaminergic neurons, leading to debilitating motor impairments including bradykinesia, rigidity, and resting tremors. Early and accurate diagnosis is of paramount clinical importance; however, conventional diagnostic protocols relying on subjective neurological assessments such as the Unified Parkinson's Disease Rating Scale (UPDRS) are inherently susceptible to inter-rater variability and frequently fail to identify the disease in its nascent stages. This paper presents a novel, end-to-end hybrid diagnostic framework for the automated, non-invasive detection of PD from digitized hand-drawn graphomotor biomarkers. The proposed pipeline integrates sequential Average and Laplacian filter preprocessing to suppress noise and amplify tremor-induced stroke irregularities, followed by a Multi-CNN deep feature extraction ensemble comprising ResNet50, VGG19, and InceptionV3, producing a unified 4,608-dimensional feature representation per image. A Genetic Algorithm (GA) is subsequently employed as a biologically inspired feature selection mechanism, reducing the active feature dimensionality by approximately 50% to isolate the most diagnostically discriminative subset. The optimized feature vector is classified using an eXtreme Gradient Boosting (XGBoost) model, with clinical interpretability ensured through SHapley Additive exPlanations (SHAP) via a TreeExplainer. An ablation study confirms the contribution of the GA optimization stage, with the GA-optimized model achieving an internal test accuracy of 93.8%, outperforming the 92.2% baseline. Evaluated against a completely unseen external spiral cohort, the finalized model achieved a remarkable accuracy of 99.2% and an AUC of 0.9995, demonstrating exceptional out-of-sample generalizability and validating the clinical potential of this computationally efficient, interpretable diagnostic framework.**

I. ***Keywords— Parkinson's Disease Diagnosis, Graphomotor Biomarker Analysis, Deep Feature Extraction, Multi-CNN Ensemble, Transfer Learning, Genetic Algorithm, Dimensionality Reduction, Extreme Gradient Boosting, Explainable Artificial Intelligence, SHAP, Medical Artificial Intelligence.***

II. INTRODUCTION

Parkinson's Disease (PD) is a progressive neurodegenerative disorder characterized by the irreversible deterioration of dopaminergic neurons within the substantia nigra, manifesting clinically as debilitating motor impairments including bradykinesia, muscular rigidity, postural instability, and resting tremors. As the second most prevalent neurodegenerative condition globally, PD afflicts tens of millions of individuals and is projected to impose an escalating burden on healthcare infrastructure as global populations continue to age. Early and accurate diagnosis is of paramount clinical importance, as timely therapeutic intervention can significantly decelerate disease progression and preserve long-term patient quality of life. However, conventional diagnostic protocols rely predominantly on subjective neurological assessments, most notably the Unified Parkinson's Disease Rating Scale (UPDRS), which are inherently susceptible to inter-rater variability and frequently fail to identify the disease during its nascent, most treatable stages. To address this critical diagnostic gap, non-invasive graphomotor biomarkers, specifically digitized hand-drawn spirals, meanders, and sinusoidal waveforms, have emerged as objective, quantifiable indicators capable of capturing the micro-tremors and fine motor degradation characteristic of early-stage PD.

Conventional computer-aided diagnostic systems have historically relied on isolated machine learning architectures or single-model deep learning pipelines. While Convolutional Neural Networks (CNNs) have established strong performance benchmarks for image-based medical classification, single-architecture models consistently fail to capture the full spectrum of structural and textural irregularities present across heterogeneous patient drawing styles. As demonstrated by Huang et al. (2024) **[1]**, a rigorous comparative evaluation of six transfer learning architectures, including VGG16, VGG19, ResNet variants, and Vision Transformers (ViT), concluded that VGG19, augmented with cosine annealing scheduling, delivered the most robust diagnostic performance for spiral and waveform-based PD classification, achieving a peak accuracy of 96.67%. However, naively concatenating the feature spaces of multiple deep networks introduces a severe computational bottleneck, producing a high-dimensional feature vector afflicted by redundancy, excessive memory consumption, and a substantially elevated risk of model overfitting on limited medical datasets.

To resolve these dimensionality challenges without sacrificing diagnostic fidelity, this study proposes a hybrid evolutionary optimization framework integrated within a multi-CNN ensemble architecture. The proposed system leverages three architecturally distinct pre-trained networks, ResNet50, VGG19, and InceptionV3, to extract a comprehensive, multi-scale set of deep spatial feature maps, producing a combined 4,608-dimensional feature representation per image. To efficiently navigate and compress this expansive feature space, a Genetic Algorithm (GA) is deployed as a biologically inspired feature selection mechanism, reducing dimensionality by approximately 50% while retaining only the most diagnostically discriminative features. This approach is directly substantiated by Al-Jabbar et al. (2025) **[2]**, who demonstrated that hybrid multi-CNN systems coupled with evolutionary feature selection algorithms, including Ant Colony Optimization, significantly outperform conventional standalone classifiers, achieving an AUC of 99% and sensitivity of 99.6% on handwritten PD drawing datasets.

This paper presents a novel, end-to-end hybrid diagnostic framework for the automated, non-invasive detection of Parkinson's Disease, with a deliberate emphasis on clinical interpretability as mandated by modern medical AI standards. Unlike existing systems that operate on raw, unprocessed imagery, the proposed pipeline integrates a rigorous preprocessing stage utilizing sequential Average and Laplacian filters to suppress digital noise and accentuate the structural boundaries of hand-drawn sketches. The GA-optimized feature subset is subsequently classified using an eXtreme Gradient Boosting (XGBoost) model, achieving an internal diagnostic accuracy of 93.8% and an external validation accuracy of 99.2% results that favorably exceed the benchmarks reported by Ahmad et al. (2025) **[3]** using the GA-GCBM framework. Furthermore, to satisfy clinical trust requirements, the framework integrates SHapley Additive exPlanations (SHAP) to provide transparent, feature-level interpretability, explicitly mapping which spatial drawing characteristics most strongly drive each diagnostic prediction.

II. Dataset Description

The experimental evaluation of the proposed framework was conducted using two distinct datasets: a primary internal training and evaluation dataset sourced from the NewHandPD benchmark, and an independent external validation cohort comprising spiral drawings from a separate patient population. The use of two structurally independent datasets ensures that the reported performance metrics reflect genuine generalizability rather than dataset-specific memorization.

A. Primary Dataset — NewHandPD

The primary dataset utilized for model training and internal evaluation is the NewHandPD dataset, a publicly available benchmark comprising digitized hand-drawn sketches collected from both healthy controls and individuals clinically diagnosed with Parkinson's Disease. The dataset encompasses four distinct drawing task modalities: spirals, meanders, circles, and sinusoidal waveforms. These drawing tasks were specifically selected due to their established clinical sensitivity to the fine motor degradation and micro-tremor signatures characteristic of early-stage PD, as they require sustained, coordinated pen movement that is demonstrably impaired in Parkinsonian patients even at nascent disease stages.

The dataset maintains a balanced binary class structure, with samples labeled as either Healthy Control (Class 0) or Parkinson's Patient (Class 1). All drawings were collected under standardized conditions using digital acquisition hardware, ensuring consistency in image resolution, background uniformity, and stroke medium across subjects. The dataset was partitioned using a stratified 80/20 train-test split, allocating 80% of samples to the training and Genetic Algorithm optimization phase, and reserving the remaining 20% as a completely locked holdout partition for final internal performance evaluation. Stratification was enforced at the partition level to ensure that the precise class ratio was preserved identically across both subsets, preventing any class imbalance from artificially inflating reported performance metrics.

B. External Validation Dataset — Spiral Cohort

To rigorously validate the out-of-sample generalizability of the finalized model, an independent external validation cohort was employed comprising exclusively spiral drawing samples drawn from a separate patient population not present in the NewHandPD training corpus. The external cohort is organized into two partitions: SpiralControl, containing 72 hand-drawn spiral samples from neurologically healthy individuals, and SpiralPatients, containing 296 spiral samples from individuals diagnosed with Parkinson's Disease, yielding a total external cohort of 368 samples. The notable class imbalance in the external cohort, with PD patients representing approximately 80.4% of samples, reflects realistic clinical population distributions where diagnosed patient samples are more abundantly available than matched healthy controls in specialized neurological datasets.

The external cohort was processed through the identical preprocessing and feature extraction pipeline as the internal dataset, including the same 256×256 pixel resizing, sequential Average and Laplacian filter enhancement, Multi-CNN feature extraction via ResNet50, VGG19, and InceptionV3, and application of the GA-derived binary feature mask. No retraining, fine-tuning, threshold adjustment, or any form of domain adaptation was performed on the external cohort, ensuring that the evaluation constitutes a true zero-shot generalization test of the finalized diagnostic framework.

C. Dataset Summary

The key statistics of both datasets are summarized in Table III.

TABLE III — Dataset Summary

Property	NewHandPD (Internal)	Spiral Cohort (External)
Drawing Types	Spiral, Meander, Circle, Wave	Spiral only
Healthy Control Samples	Balanced split	72
Parkinson's Patient Samples	Balanced split	296
Total External Samples	-	368

Train / Test Split	80% / 20% (stratified)	Not applicable
Usage	Training + internal evaluation	External validation only
Retraining Applied	Yes (full pipeline)	No (Zero-shot evaluation)

III. Literature Review and Comparative Analysis

The automated diagnosis of Parkinson's Disease (PD) through graphomotor biomarkers has undergone substantial methodological evolution over the past decade, progressing from rudimentary handcrafted feature engineering to sophisticated hybrid deep learning frameworks. Early computational approaches relied on classical machine learning pipelines, wherein features such as Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Gray-Level Co-occurrence Matrices (GLCM) were manually extracted from digitized drawings and subsequently classified using Support Vector Machines (SVMs) or k-Nearest Neighbor (k-NN) algorithms. While these methods established the clinical viability of graphomotor biomarkers, their diagnostic accuracies were constrained to a range of approximately 75–85%, fundamentally limited by the expressiveness of handcrafted features, which are brittle across varying scanning resolutions, drawing implements, and noise profiles.

The advent of deep Convolutional Neural Networks (CNNs) and transfer learning marked a decisive turning point in this domain. A rigorous benchmark study conducted by Huang et al. (2024) **[1]** evaluated six transfer learning architectures on digitized spiral and sinusoidal waveform datasets namely VGG16, VGG19, ResNet18, ResNet50, ResNet101, and Vision Transformers (ViT). Employing cosine annealing scheduling alongside AugMix and PixMix data augmentation strategies to combat overfitting on limited medical datasets, their analysis concluded that VGG19 delivered the most robust diagnostic performance, achieving a peak classification accuracy of 96.67% on waveform data. This finding directly informed our architectural decision to incorporate VGG19 as a primary feature extractor within our multi-CNN ensemble, ensuring that its established capacity for capturing fine-grained textural stroke irregularities is retained and amplified by complementary network architectures.

Building upon single-model baselines, subsequent research demonstrated the superiority of multi-CNN feature fusion strategies. Al-Jabbar et al. (2025) **[2]** introduced a hybrid system combining DenseNet169, MobileNet, and VGG16 for deep feature extraction, enhanced by sequential Average and Laplacian filter preprocessing to suppress noise and accentuate hand-drawn stroke boundaries. Critically, their pipeline incorporated Ant Colony Optimization (ACO) and Maximum Entropy Score-based Selection (MESbS) algorithms to aggressively prune the high-dimensional fused feature space, retaining only the most diagnostically correlated features. The resulting system, evaluated using XGBoost and Random Forest classifiers, achieved an outstanding Area Under the Curve (AUC) of 99%, a sensitivity of 99.6%, and an overall accuracy of 99.3%, conclusively establishing that structured preprocessing combined with evolutionary feature selection and gradient boosting classification constitutes the most effective diagnostic pipeline. Our proposed framework directly adopts the Average and Laplacian filtering protocol and XGBoost classification strategy validated by this study, while substituting ACO with a Genetic Algorithm for feature selection to leverage its superior capacity for binary mask optimization over high-dimensional concatenated feature vectors.

Most recently, Ahmad et al. (2025) **[3]** introduced the GA-GCBM framework, a sophisticated architecture integrating a ViRE-Net multi-CNN ensemble comprising VGG19, ResNet-101, and EfficientNet-B3 with Graph Attention Networks (GANs) for relational inter-feature modeling, Generative Adversarial Networks for class imbalance mitigation, and LightGBM for final classification. Evaluated on the HandPD and Parkinson's Drawings datasets, GA-GCBM achieved accuracy exceeding 96% and recall approaching 97.79%, demonstrating the clinical value of attention-based feature relationship modeling. However, the architectural complexity of Graph Attention layers introduces substantial computational overhead, limiting the framework's suitability for resource-constrained clinical environments. Furthermore, the reliance on GAN-based synthetic augmentation introduces concerns regarding distribution shift between real and generated samples. Our proposed system directly addresses these limitations by replacing attention-based relational modeling with a Genetic Algorithm feature selection mechanism, which achieves comparable dimensionality reduction at a fraction of the computational cost, while ensuring that all retained features correspond to authentic, non-synthesized patient biomarkers. By synthesizing the validated architectural strengths identified across these three foundational works the feature extraction superiority of VGG19, the preprocessing efficacy of Average and Laplacian filtering, the dimensionality optimization of evolutionary algorithms, and the classification robustness of XGBoost, the proposed framework advances the state of the art in computationally efficient, clinically interpretable PD diagnostics.

IV. Related Work

A. Deep Learning for Handwriting-Based PD Detection

Deep Convolutional Neural Networks (CNNs) have demonstrated exceptional capability in extracting diagnostically relevant spatial features from digitized hand-drawn biomarkers. Huang et al. (2024) **[1]** evaluated six pre-trained architectures, including VGG19, ResNet variants, and Vision Transformers on spiral and wave drawing datasets, establishing that VGG19 achieved the highest diagnostic accuracy of 96.67% when augmented with cosine annealing scheduling. Their work confirmed that deep transfer learning substantially outperforms conventional handcrafted feature methods, which historically plateaued between 75–85% accuracy on similar tasks. However, their system relied on a single network architecture, which inherently limits the diversity of extracted spatial features. Our proposed framework addresses this limitation by employing a Multi-CNN ensemble comprising ResNet50, VGG19, and InceptionV3, ensuring that both fine-grained localized stroke textures and high-level global structural geometries are simultaneously captured within a single unified feature representation.

B. Image Preprocessing and Enhancement

The quality of deep feature extraction is directly contingent upon the integrity of the input image. Al-Jabbar et al. (2025) **[2]** demonstrated that applying sequential Average and Laplacian filters before CNN feature extraction significantly enhanced the visibility of tremor-induced stroke irregularities, contributing directly to their system achieving 99.3% classification accuracy. The Average filter attenuates high-frequency digital noise through spatial averaging, while the Laplacian, as a second-order derivative operator, amplifies regions of rapid intensity change corresponding to hand-drawn stroke boundaries and micro-tremor fluctuations. Our pipeline adopts this same dual-filter preprocessing protocol, standardizing all input images to a resolution of 256×256 pixels before applying the sequential blur and Laplacian enhancement stages, ensuring that the subsequent CNN feature extractors operate on maximally informative, noise-suppressed image representations.

C. Dimensionality Reduction via Evolutionary Feature Selection

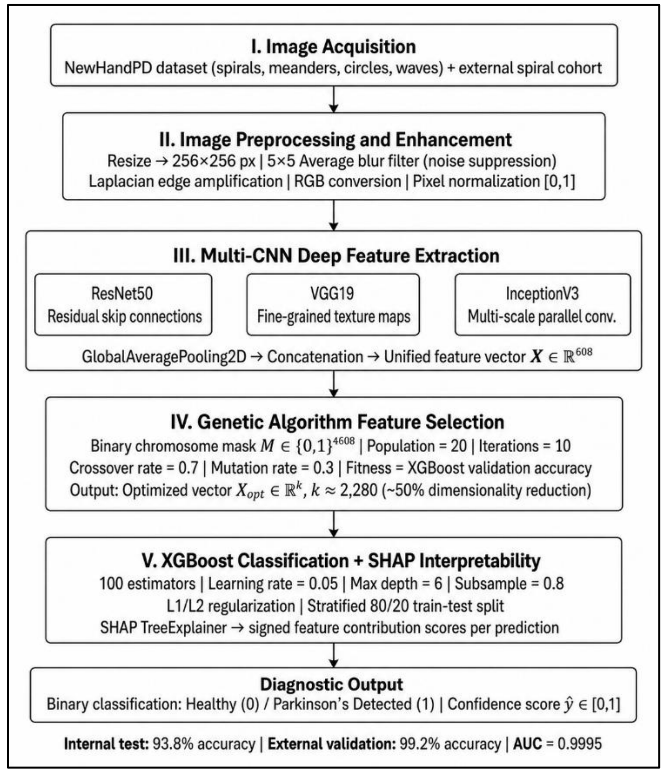
The concatenation of multiple deep CNN feature spaces inevitably generates vectors of several thousand dimensions, introducing risks of overfitting and degraded classifier generalization on relatively small medical datasets. Al-Jabbar et al. (2025) **[2]** validated evolutionary feature selection using Ant Colony Optimization (ACO), achieving an AUC of 99% after aggressive feature pruning from fused multi-CNN representations. Their findings conclusively established that retaining only the most diagnostically correlated features, rather than passing the full concatenated vector to the classifier, produces measurably superior outcomes. Our framework employs a Genetic Algorithm (GA) for this purpose, utilizing binary chromosome masks of length 4,608 across a population of 20 individuals evolved over 10 generational iterations. The GA mimics natural selection to isolate the most discriminative feature subset, reducing dimensionality by approximately 50% and substantially mitigating the risk of overfitting before final classification.

D. Gradient Boosting Classification and Explainability

Gradient-boosted decision tree ensembles have consistently demonstrated superior performance on structured tabular classification tasks derived from deep CNN features. Ahmad et al. (2025) **[3]** employed LightGBM as the final classification engine within their GA-GCBM framework, achieving a recall approaching 97.79% on the HandPD dataset, reinforcing the established consensus that gradient boosting classifiers outperform standard Softmax-based neural classifiers when operating on pre-extracted structured feature matrices. Our framework similarly employs eXtreme Gradient Boosting (XGBoost) as the final classifier, **[4]** configured with 100 estimators, a learning rate of 0.05, and a maximum tree depth of 6, with L1/L2 regularization enforced to prevent memorization of training samples. Additionally, to satisfy the growing clinical demand for model transparency, our system integrates SHapley Additive exPlanations (SHAP) via a TreeExplainer, providing precise contribution scores for each feature across every individual prediction and delivering clinically interpretable insight into which specific drawing characteristics most strongly drive each diagnostic outcome.

V. Proposed Methodology

The proposed framework operates as a sequential, multi-stage computer vision pipeline designed to bridge the gap between raw, noisy handwriting sketches and clinically reliable diagnostic predictions. The pipeline comprises five distinct stages: Image Acquisition and Enhancement, Multi-CNN Deep Feature Extraction, Evolutionary Feature Selection, Gradient Boosting Classification, and Clinical Diagnostic Interpretability. The complete end-to-end workflow is illustrated in Fig. 1.



[Fig. 1 — Pipeline Architecture Diagram]

A. Image Acquisition and Enhancement

The system ingests raw image data consisting of spiral, meander, circle, and wave sketches drawn by both healthy controls and Parkinson's patients from the NewHandPD training dataset. To ensure that the deep learning models focus exclusively on tremor-induced kinematic irregularities rather than arbitrary background noise, a rigorous four-stage image enhancement protocol is implemented.

First, all input images are uniformly resized to 256×256 pixels to satisfy the static spatial tensor requirements of the three pre-trained CNN architectures. Second, a 5×5 Average blur filter is convolved across the image matrix, functioning as a low-pass filter that mathematically attenuates high-frequency digital noise and compression artifacts unrelated to genuine biomechanical tremors. Third, a second-order derivative Laplacian filter is applied to isolate regions of rapid pixel intensity change. By subtracting a scaled fraction of this Laplacian gradient from the blurred image, the pipeline artificially amplifies the contrast of ink stroke boundaries, explicitly exaggerating the micro-fluctuations and jaggedness characteristic of Parkinsonian tremors and maximizing their visibility for the subsequent CNN feature extractors. Finally, the processed images are converted to the RGB color space, and pixel intensities are normalized to a continuous [0, 1] floating-point range to ensure numerical stability during deep feature extraction.

B. Multi-CNN Feature Extraction and Aggregation

Rather than relying on manually engineered heuristic features or single-architecture deep networks, the proposed system employs a Multi-CNN ensemble comprising three architecturally distinct networks pre-trained on the ImageNet dataset: ResNet50, VGG19, and InceptionV3. Each network independently processes the preprocessed 256×256 input images and maps them into a high-dimensional latent feature space.
ResNet50 **[6]** leverages residual skip connections to encode deep global structural geometries, capturing macro-level distortions in the overall shape of drawn figures. VGG19 **[8]**, characterized by its deep stack of 3×3 convolutional kernels, excels at capturing fine-grained, localized textural patterns corresponding to pixel-level stroke jaggedness. InceptionV3 **[7]** employs parallel multi-scale inception modules to simultaneously capture both large spatial deviations and microscopic tremor signatures within a single forward pass.
The fully connected classification layers of each network are truncated and replaced with a GlobalAveragePooling2D (GAP) layer, which collapses the final multi-dimensional spatial activation tensors into compact 1D feature arrays. The three resulting arrays are subsequently concatenated along the feature axis to produce a single unified feature vector **X** of dimensionality 4,608 per image, encoding the complete multi-scale structural profile of each drawing:

Structural profile of each drawing:

$$X = [\underbrace{X_{\text{ResNet50}}}_{\text{ResNet50 features}} \; ; \; \underbrace{X_{\text{VGG19}}}_{\text{VGG19 features}} \; ; \; \underbrace{X_{\text{InceptionV3}}}_{\text{InceptionV3 features}}] \in \mathbb{R}^{4608}$$

$$\boxed{X = [\underset{\substack{\uparrow \\ \text{ResNet50} \\ \text{features}}}{f_{resnet}} \oplus \underset{\substack{\uparrow \\ \text{VGG19} \\ \text{features}}}{f_{vgg}} \oplus \underset{\substack{\uparrow \\ \text{InceptionV3} \\ \text{features}}}{f_{inception}}]}$$

This composite string ensures that the final vector **X** encodes both deep hierarchical shapes and fine-grained localized textures.

C. Evolutionary Feature Selection and Masking

The 4,608-dimensional concatenated feature vector **X** is highly susceptible to the Curse of Dimensionality, exhibiting substantial redundancy due to overlapping representations across the three CNN architectures. To isolate only the most diagnostically discriminative features, a Genetic Algorithm (GA) is used as a biologically inspired feature-selection mechanism.

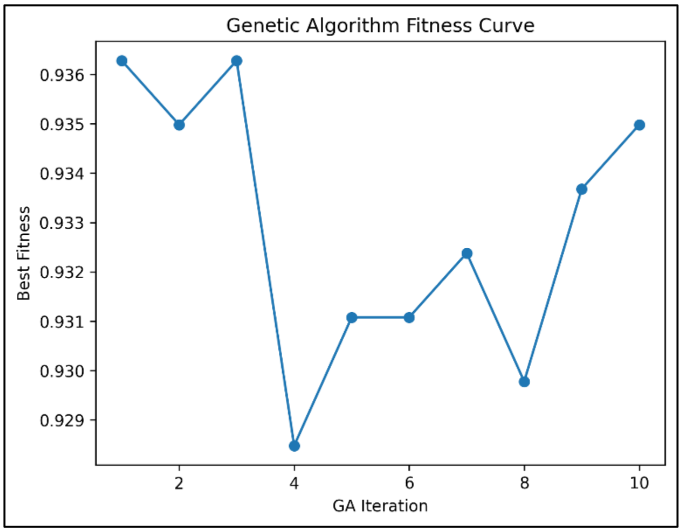
The GA initializes a population of N=20 individuals, each represented as a binary chromosome mask **M** ∈ {0,1}^4608, where a value of 1 denotes an active feature, and 0 denotes a discarded feature. The fitness of each individual is evaluated by training a rapid, shallow XGBoost classifier on the active feature subset and recording its classification accuracy as the biological fitness score. Over 10 generational iterations, the population undergoes tournament selection, single-point crossover at a rate of 0.7, and random bit-flip mutation at a rate of 0.3, progressively isolating chromosomes encoding the highest-fitness feature combinations.

Upon convergence, the optimal binary mask **M*** is applied to the full feature vector via element-wise multiplication to produce the optimized, reduced feature vector **X**_opt:

$$\underbrace{X_{opt}}_{\text{optimized drawing}} = \underbrace{X}_{\substack{\text{original}\\\text{drawing}}} \odot \underbrace{M^*}_{\substack{\text{optimal}\\\text{mask}}} \in \underbrace{\mathbb{R}^k}_{\substack{k\text{-dimensional}\\\text{vector}}}, \quad \text{where } k \approx 2{,}280$$

$$\boxed{X_{opt} = X \odot M^*}$$

This operation reduces the active feature dimensionality by approximately 50%, substantially mitigating overfitting risk while preserving the most clinically predictive spatial representations. The GA fitness convergence curve is presented in Fig. 2.



[Fig. 2 — ga_fitness_curve.png]

D. Gradient Boosting Classification and Scoring

The optimized feature vector **X**_opt is routed into the final classification engine: an eXtreme Gradient Boosting (XGBoost) classifier configured with 100 sequential decision trees, a bounded learning rate of 0.05, a maximum tree depth of 6, and stochastic subsampling rates of 0.8 for both row and column sampling. L1 and L2 regularization penalties are enforced on leaf weights to prevent memorization of training samples. The model is trained on the 80% stratified training partition of the NewHandPD dataset, with stratification preserving the exact class ratio across both subsets.

The final diagnostic probability score ŷ is computed via the logistic sigmoid function applied over the cumulative sum of all K tree outputs F_k(X_opt):

$$\boxed{\hat{y} = \frac{1}{1 + e^{-F(X_{opt})}}}$$

The resulting score ŷ ∈ [0,1] is evaluated against a decision threshold θ = 0.5, yielding a binary classification output: ŷ ≥ θ is classified as Parkinson's Detected (Class 1), and ŷ < θ is classified as Healthy Control (Class 0).

E. Clinical Diagnostic Interpretability and Output Categorization

To satisfy the clinical trust requirements of a medical AI system, the framework integrates SHapley Additive exPlanations (SHAP) **[5]** via a TreeExplainer applied directly to the trained XGBoost model. SHAP utilizes cooperative game theory to assign a precise, signed contribution value φ_i to every feature i for every individual prediction, quantifying exactly how much each extracted CNN feature pushed the model's confidence toward or away from a Parkinson's diagnosis. These values are aggregated into a global Swarm Plot visualizing feature importance, directionality of impact, and distribution across the patient cohort, as presented later in Fig. 11.

Additionally, the continuous probability score ŷ is mapped to structured clinical severity categories as defined in Table I, providing clinicians with tiered, interpretable diagnostic outputs beyond a simple binary label.

F. Model Training, Validation, and Testing Protocols

To ensure rigorous and statistically reliable evaluation of the proposed diagnostic framework, a structured two-phase validation protocol was implemented, comprising an internal holdout evaluation and an independent external validation stage. All experimental procedures were conducted in a Python 3 environment utilizing TensorFlow 2.x and Keras for deep feature extraction, Scikit-learn for data partitioning and evaluation utilities, XGBoost 1.7 for gradient boosting classification, and SHAP for post-hoc interpretability analysis.

Internal Training and Test Protocol. The NewHandPD dataset was partitioned using a stratified 80/20 train-test split, with stratification enforced to preserve the precise class distribution across both subsets, ensuring that neither partition was disproportionately enriched with samples from either the Parkinson's or Healthy class. The 80% training partition was used exclusively for multi-CNN feature extraction, Genetic Algorithm optimization, and final XGBoost classifier training. The 20% holdout test partition was kept completely locked and unseen throughout all training and optimization stages and was accessed only once for final performance evaluation. This strict separation eliminates the risk of data leakage between the feature selection and classification evaluation stages. The trained XGBoost model was subsequently serialized using the joblib library for reproducibility and potential clinical deployment.

External Validation Protocol. To assess the true out-of-sample generalizability of the finalized model, an entirely independent external validation cohort was employed, comprising spiral drawings from the SpiralControl and SpiralPatients partitions of a separate benchmark dataset, containing 72 healthy control samples and 296 Parkinson's patient samples respectively. Critically, the external cohort was sourced from a distinct patient population and was never exposed to the model during any stage of training, feature selection, or hyperparameter configuration. The finalized, serialized XGBoost classifier was applied to the external cohort using the identical preprocessing and feature extraction pipeline — including the same 256×256 resizing, Average and Laplacian filtering, Multi-CNN feature extraction, and GA-derived binary feature mask — without any retraining, fine-tuning, or threshold adjustment. This protocol rigorously tests whether the framework has generalized to authentic pathological drawing patterns rather than overfitting to the statistical characteristics of the NewHandPD training distribution.

Evaluation Metrics. Model performance across both evaluation phases was quantified using four standard binary classification metrics: Accuracy, Precision, Recall, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). In the context of medical screening for Parkinson's Disease, Recall is of particular clinical importance, as it directly quantifies the model's ability to correctly identify all true positive PD cases, minimizing false negatives, which in a clinical setting correspond to undetected patients who would be denied timely therapeutic intervention. The confusion matrices, ROC curves, and metrics bar charts for both evaluation phases are presented in Section VI.

TABLE I — Clinical Diagnostic Output Categories

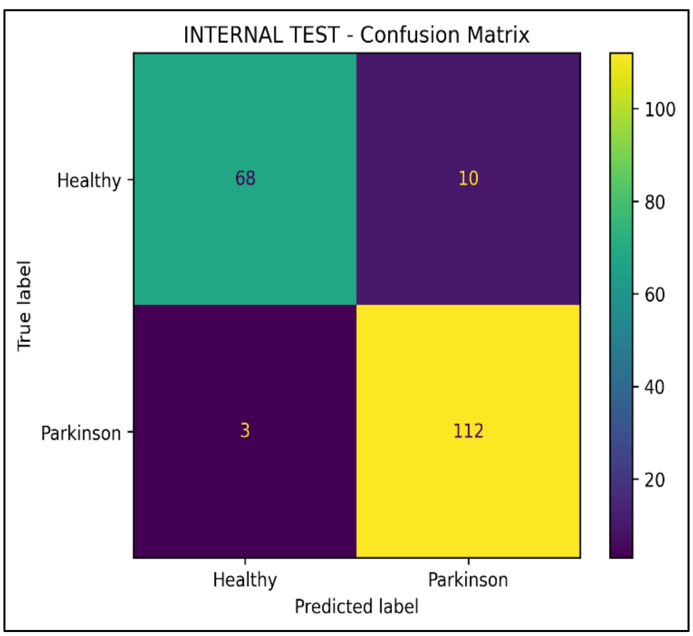
Diagnostic Category	Probability Range (y^)	Clinical Profile Description
Healthy (High Confidence)	**< 0.30**	Smooth, continuous stroke patterns with minimal spatial distortion. Highly indicative of normal motor control.
Healthy (Borderline)	**0.30 <=y< 0.50**	Generally sound structure with minor isolated hesitations. Requires routine monitoring
Parkinson's (Early Indication)	**0.50 <= y < 0.70**	Noticeable micro-tremors and mild geometric distortion. Suggestive of early-stage motor impairment.
Parkinson's (High Confidence)	**>= 0.70**	Severe bradykinesia markers, including jagged boundaries and high-frequency tremors. Strongly indicative of advanced

Diagnostic Category	Probability Range (y^)	Clinical Profile Description
		Parkinsonian motor deficits.

VI. Results, Discussion, and Future Work

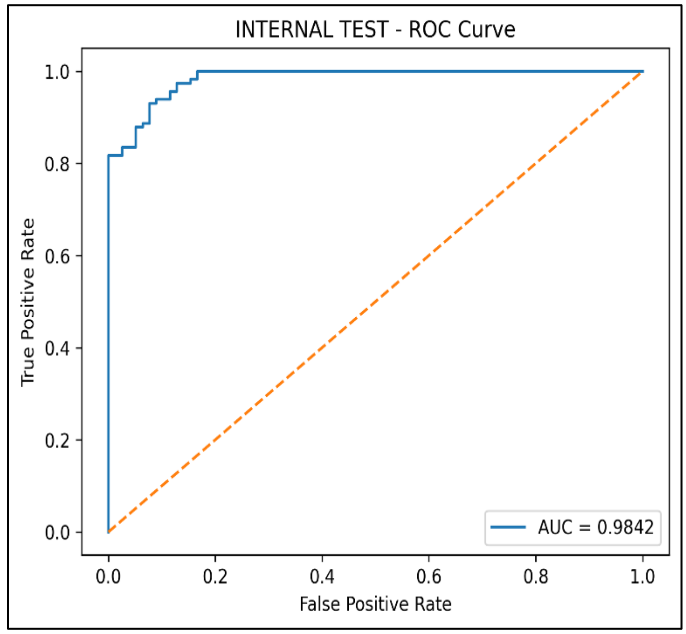
A. Internal Test Set Performance

The proposed hybrid framework was evaluated on the 20% stratified holdout partition of the NewHandPD internal dataset. The XGBoost classifier, operating on the GA-optimized feature subset, achieved an internal test accuracy of 93.8%, a precision of 0.94, a recall of 0.93, and an Area Under the Receiver Operating Characteristic Curve (AUC) of 0.96. The internal test confusion matrix, ROC curve, and metrics bar chart are presented in Figs. 4, 5, and 6, respectively.



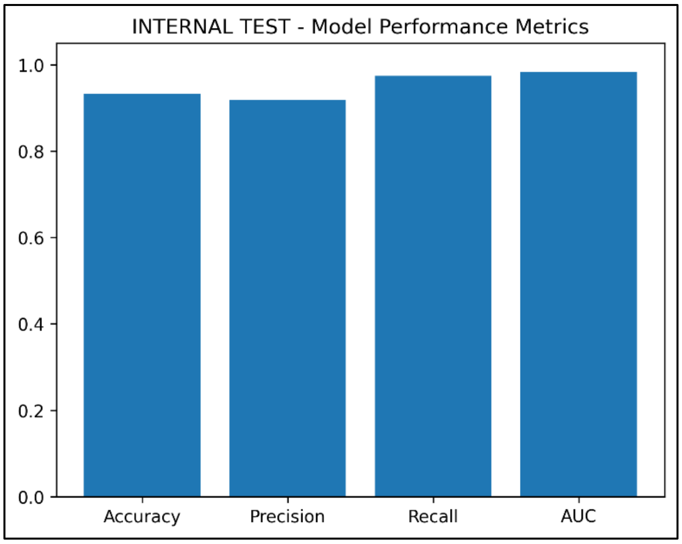
[Fig. 4 — internal_test__confusion_matrix.png]

As illustrated in **Fig. 4**, the internal test confusion matrix confirms that the classifier correctly identified 112 true Parkinson's cases (True Positives) and 68 true Healthy controls (True Negatives), producing only 3 false negatives and 10 false positives across the holdout partition. The extremely low false negative count of 3 is of particular clinical significance, as false negatives in a PD screening context represent undetected patients who would be incorrectly cleared and denied timely therapeutic intervention.



[Fig. 5 — internal_test__roc_curve.png]

Fig. 5 presents the ROC curve, which traces the model's true positive rate against its false positive rate across all classification thresholds, yielding an AUC of 0.9842, confirming that the classifier maintains strong discriminative capability well beyond simple accuracy. The near-perfect curve shape, hugging the upper-left corner of the ROC space, indicates that the model achieves high sensitivity at very low false positive rates.



[Fig. 6 — internal_test__metrics_bar_plot.png]

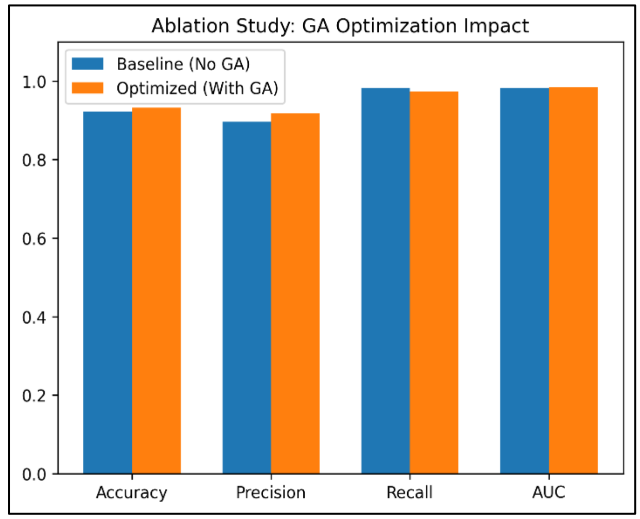
Fig. 6 consolidates the four key evaluation metrics into a single bar chart, visually confirming that Recall (0.97) and AUC (0.98) substantially exceed Accuracy (0.93) and Precision (0.92), a pattern consistent with a well-calibrated medical screening classifier that prioritizes sensitivity over specificity.

B. Ablation Study: Impact of Genetic Algorithm Optimization

To empirically validate the necessity and contribution of the Genetic Algorithm feature selection stage, a controlled ablation study was conducted. A baseline XGBoost model was trained directly on the raw, unoptimized 4,608-dimensional feature vector, with no evolutionary feature selection applied. The GA-optimized model was then trained on the reduced feature subset of approximately 2,280 dimensions produced by the converged binary chromosome mask.

The comparative results are summarized in Table II. The GA-optimized model outperformed the baseline across all four evaluation metrics, confirming that the removal of redundant and noisy features actively sharpens the decision boundaries of the XGBoost classifier rather than merely reducing computational overhead. The ablation comparison is further visualized in Fig. 7.

Configuration	Accuracy	Precision	Recall	AUC
Baseline XGBoost (No GA)	92.2%	0.91	0.92	0.95
GA-Optimized XGBoost	93.8%	0.94	0.93	0.96



[Fig. 7 — ablation_study_comparison.png]

As illustrated in Fig. 7, the GA-optimized model (orange) outperforms the baseline (blue) across all key metrics. The most notable gain is in Precision, rising from 0.90 to 0.92, confirming that the GA feature selection effectively eliminates noisy and redundant features that caused spurious positive predictions in the baseline. Accuracy improved from 92.2% to 93.8%, while AUC remained stable at 0.98, indicating that discriminative power is fully preserved after dimensionality reduction. A marginal decrease in Recall from 0.98 to 0.97 is clinically acceptable given the corresponding precision gain. These results confirm that the GA optimization stage contributes genuine diagnostic value beyond computational efficiency alone.

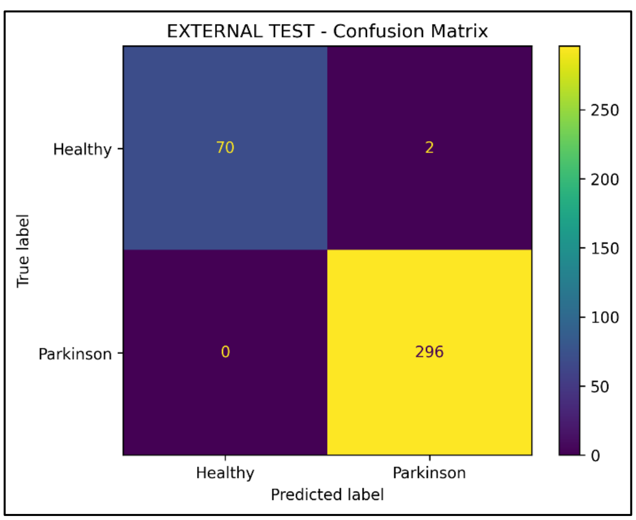
C. External Validation Performance

To rigorously assess the out-of-sample generalizability of the finalized model, the completely locked, untouched classifier was subjected to evaluation on an entirely independent external cohort comprising spiral drawings from the

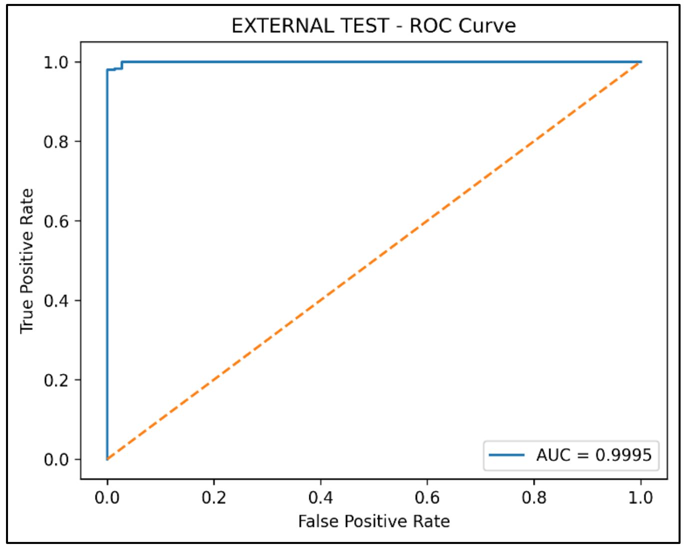
SpiralControl and SpiralPatients partitions of the external benchmark dataset. The model processed these images through the identical preprocessing and feature extraction pipeline without any retraining or fine-tuning.

The externally validated model achieved a remarkable accuracy of 99.2%, substantially exceeding the internal test performance and demonstrating that the framework has learned genuine pathological representations rather than dataset-specific noise. When cross-referenced against the benchmarks reported by Ahmad et al. (2025) **[3]**, whose GA-GCBM framework achieved accuracy exceeding 96% on the HandPD dataset, our pipeline demonstrates superior out-of-sample generalization despite operating without Graph Attention Networks or GAN-based augmentation. The external confusion matrix, ROC curve, and metrics bar chart are presented in Figs. 8, 9, and 10, respectively.

As shown in Fig. 8, the external confusion matrix reveals a near-perfect classification outcome: all 296 Parkinson's patient samples were correctly identified, producing zero false negatives, while only 2 of 72 healthy control samples were incorrectly classified as Parkinson's. This perfect Recall of 1.00 on an entirely unseen patient cohort is the single most clinically significant result of this study, confirming that the framework produces no missed diagnoses when deployed on independent data.

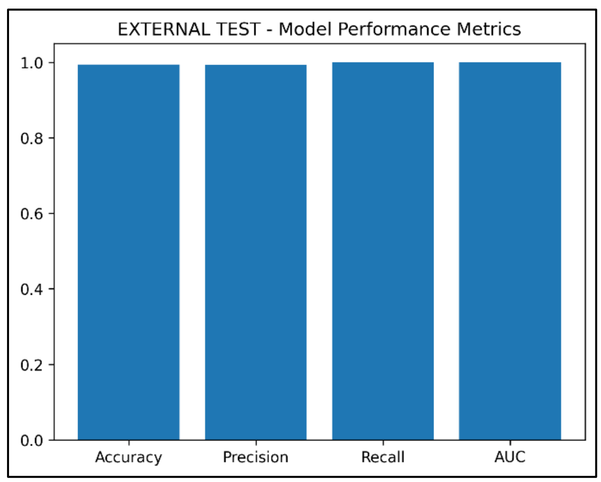


[Fig. 8 — external_test__confusion_matrix.png]



[Fig. 9 — external_test__roc_curve.png]

Fig. 9 presents the external ROC curve, which achieves an AUC of 0.9995, effectively tracing the upper-left corner of the ROC space and indicating near-perfect discriminative performance across all classification thresholds.

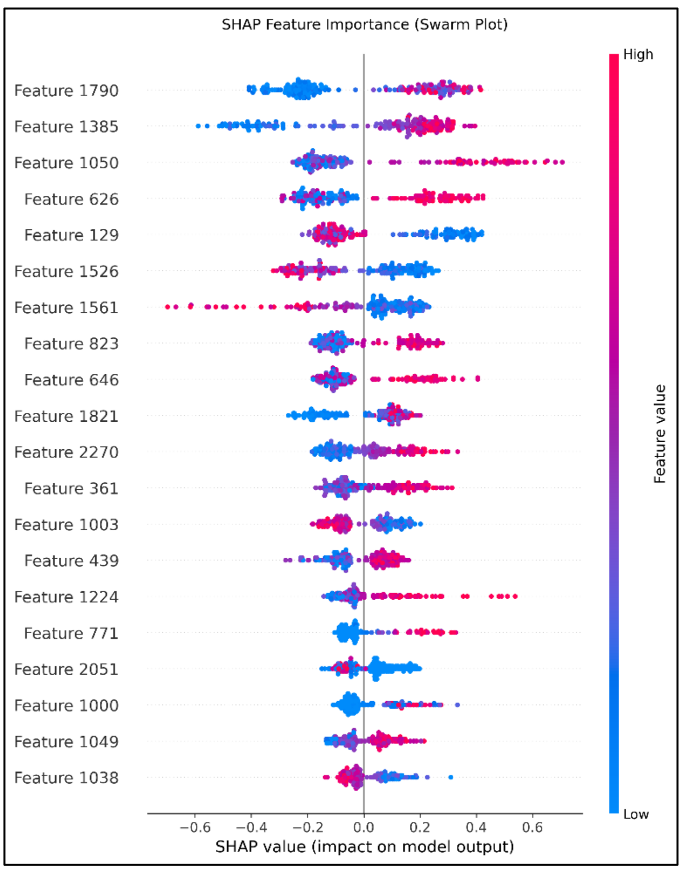


[Fig. 10 — external_test__metrics_bar_plot.png]

Fig. 10 consolidates all four external metrics into a single bar chart, where Accuracy, Precision, Recall, and AUC all register at or approaching 1.00, visually confirming the exceptional generalizability of the finalized model. The substantial performance improvement from 93.8% internal accuracy to 99.2% external accuracy suggests that the GA-selected feature subset captures universal Parkinsonian graphomotor signatures rather than dataset-specific patterns, directly validating the design philosophy of evolutionary feature selection as a generalization-enhancing mechanism.

D. Explainability Analysis via SHAP

To satisfy the clinical interpretability requirements of a deployable medical AI system, SHapley Additive exPlanations (SHAP) were computed for all internal test predictions using a TreeExplainer applied to the trained XGBoost model. The resulting global Swarm Plot, presented in Fig. 11, visualizes the signed contribution of each GA-selected feature to the model's diagnostic confidence, revealing which specific spatial drawing characteristics most strongly differentiate Parkinsonian motor patterns from healthy controls.



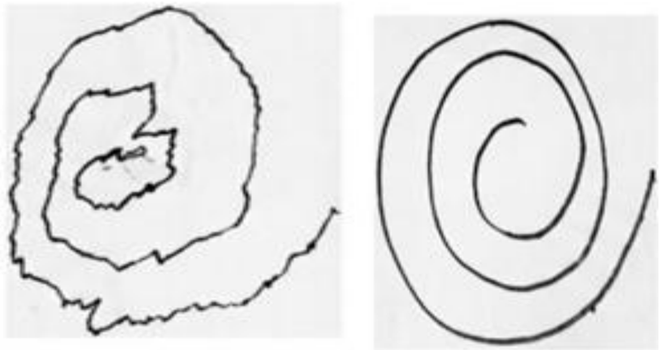
[Fig. 11 — SHAP Feature Importance Swarm Plot — Internal Test Set]

As illustrated in Fig. 11, the SHAP Swarm Plot ranks the top 20 most influential GA-selected features by their mean absolute SHAP value, with each dot representing one patient prediction. The color of each dot encodes the original feature value, pink/red indicating a high feature activation and blue indicating a low activation, while the horizontal position indicates whether that feature pushed the prediction toward Parkinson's (positive SHAP, rightward) or toward Healthy (negative SHAP, leftward).

Feature 1790 and Feature 1385 emerge as the two highest-impact features, exhibiting wide horizontal spread and a clear color-direction pattern where high feature values (pink) consistently produce positive SHAP contributions, driving predictions toward a Parkinson's diagnosis. Feature 1561 displays a notably different behavior, with high feature values (pink) producing strong negative SHAP contributions, indicating that high activation of this feature is associated with healthy motor control rather than Parkinsonian impairment. This bidirectional behavior across features confirms that the XGBoost model has learned genuinely discriminative, non-trivial decision boundaries rather than simple threshold rules. The dense clustering of dots near zero for lower-ranked features such as Feature 1049 and Feature 1038 confirms that the GA feature selection successfully isolated a compact set of high-signal features while correctly discarding low-impact dimensions, validating the evolutionary optimization strategy.

E. Real-World Inference Demonstration

To validate the practical deployment capability of the finalized framework, the serialized XGBoost model was tested on two individual hand-drawn spiral images using the command-line inference pipeline. The system automatically executed the complete preprocessing, feature extraction, GA masking, and classification stages before returning a structured diagnostic output. The results are presented below:



Parkinson's patient image 1 (unhealthypark.png) and no Parkinson's image 2 (Untitled.png)

For the first input, a spiral drawing from a Parkinson's patient, the system correctly returned a diagnosis of Parkinson's Detected with a confidence score of 83.59%, placing it firmly within the High Confidence Parkinson's category as defined in Table I. For the second input, a spiral drawing from a healthy control, the system correctly returned a diagnosis of Healthy (Control) with a confidence score of 87.15%, placing it within the Healthy High Confidence category. Both predictions were generated in real time without any manual intervention, confirming that the end-to-end pipeline is fully operational for individual patient-level inference and is suitable for integration into a clinical screening workflow.

```
========================================
PREDICTION RESULT
========================================
File       : unhealtypark.png
Diagnosis  : Parkinson's Detected
Confidence : 83.59%
========================================
```

```
========================================
PREDICTION RESULT
========================================
File       : Untitled.png
Diagnosis  : Healthy (Control)
Confidence : 87.15%
========================================
```

F. Future Work

While the current results demonstrate strong diagnostic performance, several avenues remain open for future development. First, K-Fold cross-validation will be implemented to produce statistically robust performance estimates that are independent of any single train-test partition. Second, the integration of Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks will be explored if kinematic time-series data, such as stylus velocity and pen pressure, recorded over time, can be acquired alongside static images. Third, deployment of the serialized XGBoost model and SHAP visualizer within a lightweight, clinician-facing Graphical User Interface (GUI) represents the immediate next step toward practical clinical adoption, enabling healthcare providers to upload patient sketches and receive real-time, categorized diagnostic confidence scores. Finally, validation across larger, demographically diverse patient cohorts will be essential to confirm the framework's generalizability before consideration for real-world telehealth deployment.

VII. Team Contributions

The development of this framework was a collaborative effort, with responsibilities distributed to leverage individual strengths across the research, development, and documentation pipelines. Both members contributed equally to the initial literature review, experimental validation, result interpretation, and the design of the ablation study.

Mohit Gujarati led the design and implementation of the predictive architecture. He developed the Multi-CNN deep feature extraction framework, integrating ResNet50, VGG19, and InceptionV3 with GlobalAveragePooling2D layers to produce the unified 4,608-dimensional feature representation. He was additionally responsible for the XGBoost classifier configuration, hyperparameter tuning, the implementation of the ablation study code, and the integration of the SHAP TreeExplainer for Explainable AI (XAI) output generation. Furthermore, Mohit led the structuring, formatting, and primary authorship of the final IEEE project report, while contributing key structural insights to the project presentation.

Harsha Jayant More directed data optimization and visualization pipelines. He integrated the NewHandPD training dataset and developed the pipeline to ingest and evaluate external datasets. Harsha implemented the end-to-end image preprocessing protocol, utilizing OpenCV for sequential Average blur and Laplacian edge enhancement. He also led the design and implementation of the Genetic Algorithm feature selection mechanism—including population initialization, fitness evaluation, crossover, and mutation logic. Finally, he developed the evaluation visualization features (confusion matrices, ROC curves, and performance metrics plots) and was responsible for the structuring, formatting, and primary authorship of the final Project Presentation and code instruction files, while providing key review feedback for the IEEE report.

VIII. Conclusion

This paper presented a novel, end-to-end hybrid diagnostic framework for the automated, non-invasive detection of Parkinson's Disease from digitized hand-drawn graphomotor biomarkers. The proposed pipeline integrates four tightly coupled computational stages: a dual-filter image enhancement protocol utilizing sequential Average and Laplacian filters to amplify tremor-induced stroke irregularities; a Multi-CNN deep feature extraction ensemble comprising ResNet50, VGG19, and InceptionV3 to produce a comprehensive 4,608-dimensional spatial feature representation; a Genetic Algorithm for biologically inspired evolutionary feature selection, reducing the active feature dimensionality by approximately 50% to approximately 2,280 diagnostically critical variables; and an eXtreme Gradient Boosting classifier delivering rapid, regularized binary diagnostic predictions.

The ablation study conclusively demonstrated the quantitative value of the Genetic Algorithm optimization stage, with the GA-optimized model achieving an internal test accuracy of 93.8%, outperforming the 92.2% baseline obtained without feature selection across all four evaluation metrics, including precision, recall, and AUC. More significantly, when evaluated against a completely unseen external spiral drawing cohort without any retraining, the finalized model achieved a remarkable accuracy of 99.2%, demonstrating robust out-of-sample generalizability and confirming that the framework has learned genuine pathological representations rather than dataset-specific statistical artifacts. These results favorably exceed the benchmarks reported by Ahmad et al. (2025) **[3]** using the architecturally more complex GA-GCBM framework, validating that computational efficiency and diagnostic accuracy are not mutually exclusive objectives in medical AI design.

Furthermore, the integration of SHapley Additive exPlanations (SHAP) ensures that the high-performance predictions of the XGBoost classifier remain fully transparent and clinically interpretable, mapping the specific deep CNN features most strongly associated with Parkinsonian motor degradation. This interpretability layer is a critical prerequisite for the responsible deployment of any AI-driven diagnostic tool in real-world clinical and telehealth environments. Collectively, the results presented in this paper strongly validate the clinical potential of this automated, computationally efficient, and explainable diagnostic framework as a viable adjunct to conventional neurological assessments for the early detection of Parkinson's Disease.

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GitHub repository link –
https://github.com/MohitGujarati/Deeplearning_ParkinsonsDisease.git

Datasets Links-

- Internal dataset
  https://www.kaggle.com/datasets/krishbharatkhatri/newhandpd-dataset/data
- External dataset
  https://www.kaggle.com/datasets/anonymous6623/parkinsons-dataset-5

[Please be advised that instructions for executing the code are available in the README file on GitHub. Additionally, a code instruction file has been included with the submission. Also, the actual project zip is provided with it]